The documentation and process conversion measures necessary to comply with this revision shall be completed by 2 Jan 92

INCH-POUND

MIL-S-19500/607
1 September 1992

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, FIELD EFFECT TRANSISTORS, N-CHANNEL AND P-CHANNEL, SILICON TYPE 2N7337 JANTX, JANTXV, JANS, AND JANC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for N-channel, and P-channel, enhancement-mode, MOSFET, power transistor intended for use in high density power switching applications. Three levels of product assurance are provided for each device type as specified in MIL-S-19500, with avalanche energy ratings (E_{AS} and E_{AR}) and maximum avalanche current (I_{AR}). One level of product assurance for die (element evaluation).

1.2 Physical dimensions. See figure 1 (MO-036AB), see figure 2 for JANC die dimensions.

1.3 Maximum ratings (T_A = +25°C, unless otherwise specified).

Type	P_T 1/ T_C = +25°C (free air)	V_{GS}	I_{D1} 2/ T_C = +25°C		I_{D2} 2/ T_C = +100°C		I_S	
	W	V dc	A dc		A dc		A dc	
			N-CH	P-CH	N-CH	P-CH	N-CH	P-CH
2N7337	1.4	±20	1.0	-1.0	0.6	-0.6	1.0	-1.0

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ECT, 1507 Wilmington Pike, Dayton, OH 45444-5280, using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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E_{AS}	E_{AR}	I_{AR} 2/		I_{DM} 3/		T_{op} and T_{STG}	Max $R_{DS(on)}$ 1/ V_{GS} = 10 V dc, I_D = I_{D2}				$R_{\theta JA1}$	$R_{\theta JA2}$
							ohms T_J = +25°C		ohms T_J = +150°C			
mJ	mJ	A		A (pk)		°C					°C/W maximum (1 die)	°C/W maximum (4 die)
		N-CH	P-CH	N-CH	P-CH		N-CH	P-CH	N-CH	P-CH		
75	.14	1.0	-1.0	4.0	-4.0	-55 to +150	0.7	0.7	1.4	1.4	90	50

1/ Derate linearly 1.2 W/°C for T > +25°C. $P_T = \frac{T_{J\ max} - T_C}{R_{\theta JA}}$

2/ $I_D = \frac{T_J(max) - T_C}{R_{\theta JA} \times (R_{DS(on)} \text{ at } T_J(max))}$

3/ $I_{DM} = 4\ I_{D1}$; I_{D1} as calculated in footnote 2.

1.4 Primary electrical characteristics at T_C = +25°C.

Type	Min $V_{(BR)DSS}$		$V_{GS(th)1}$				Max I_{DSS1}		Max $R_{DS(on)1}$ 1/	
	N-CH	P-CH	N-CH		P-CH		N-CH	P-CH	N-CH	P-CH
	V_{GS} = 0V I_D = 1.0 mA	V_{GS} = 0V I_D = -1.0 mA	$V_{DS} \geq V_{GS}$ I_D = 0.25 mA		$V_{DS} \geq V_{GS}$ I_D = -0.25 mA		V_{DS} = 80% of rated V_{DS}		V_{GS} = 10 V $I_D = I_{D2}$	V_{GS} = -10 V $I_D = I_{D2}$
	V dc		V dc				μA dc		ohms	
			min	max	min	max				
2N7337	100	-100	2.0	4.0	-2.0	-4.0	25	-25	0.7	0.7

1/ Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

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SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 40, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and on figure 1 herein.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-STD-750, MIL-S-19500, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic protection.

3.5.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of electrostatic charge. The following handling practices shall be followed:

a. Devices shall be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

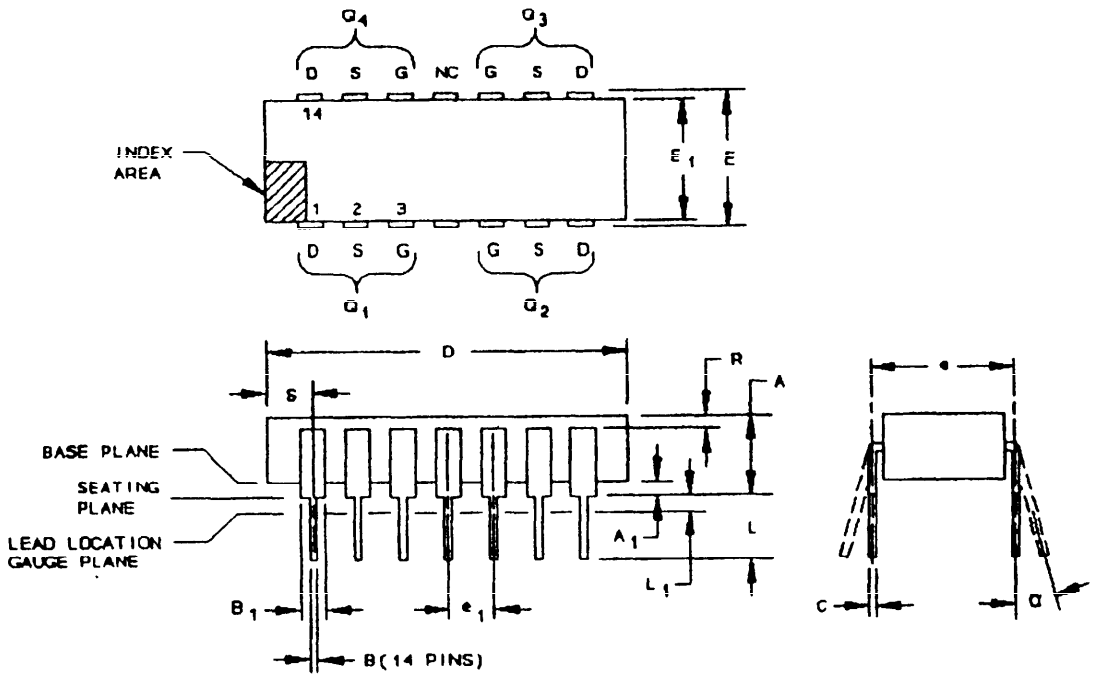
e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care shall be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq 100$ k, whenever bias voltage is to be applied drain to source.

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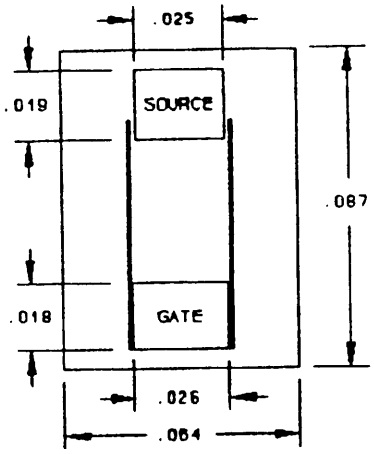


Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	0.105	0.165	2.67	4.19
A_1	0.025	0.050	0.64	1.27
B	0.015	0.021	0.381	0.533
B_1	0.042	0.058	1.07	1.47
C	0.008	0.012	0.203	0.305
D	0.735	0.760	18.67	19.30
E	0.295	0.322	7.49	8.18
E_1	0.280	0.305	7.11	7.75

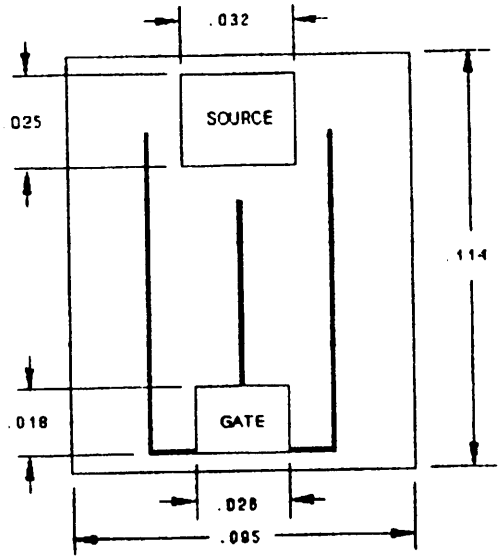
Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
e	0.290	0.315	7.37	8.00
e_1	0.095	0.105	2.41	2.67
L	0.125	0.175	3.18	4.45
L_1	0.000	0.030	0.00	0.76
α	0°	15°	0°	15°
R	0.010	---	0.25	---
S	0.030	0.095	0.76	2.41

FIGURE 1. Dimensions and configuration (MO-036AB).

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HEX-1: 100V, N-CHANNEL



HEX-2: 100V, P-CHANNEL

Inches	mm	Inches	mm
.018	0.46	.064	1.63
.019	0.48	.087	2.21
.025	0.64	.095	2.41
.026	0.67	.114	2.90
.032	0.82		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
4. The physical characteristics of the die thickness are .0187 (0.474 mm). The metals are chromium, nickel, and silver. The top metal is aluminum and the back contact is the drain.

FIGURE 2. JANC A die dimensions.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500, and as specified herein. Alternate flow is allowed for qualification inspection in accordance with figure 2 of MIL-S-19500.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with MIL-S-19500, and table IV herein.

4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
1/ 2/	Gate stress test (see 4.5.5)	Gate stress test (see 4.5.5)
1/	$Z_{\theta JC}$ Method 3161 (see 4.5.3)	$Z_{\theta JC}$ Method 3161 (see 4.5.3)
9 1/	I_{GSS1}, I_{DSS1}, subgroup 2 of table I herein	Subgroup 2 of table I herein
10	Method 1042, test condition B	Method 1042, test condition B
11	I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$ Subgroup 2 of table I herein; ΔI_{GSS1} = ±20 nA dc or ±100% of initial value, whichever is greater ΔI_{DSS1} = ±25 µA dc or ±100% of initial value, whichever is greater	I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$ Subgroup 2 of table I herein
12	Method 1042, test condition A,	Method 1042, test condition A, optional t = 48 hours at 175°C
13	Subgroups 2 and 3 of table I herein; ΔI_{GSS1} = ±20 nA dc or ±100% of initial value, whichever is greater ΔI_{DSS1} = ±25 µA dc or ±100% of initial value, whichever is greater $\Delta R_{DS(on)1}$ = ±20% of initial value $\Delta V_{GS(th)1}$ = ±20% of initial value	Subgroup 2 of table I herein; ΔI_{GSS1} = ±20 nA dc or ±100% of initial value, whichever is greater ΔI_{DSS1} = ±25 µA dc or ±100% of initial value, whichever is greater $\Delta R_{DS(on)1}$ = ±20% of initial value $\Delta V_{GS(th)1}$ = ±20% of initial value

1/ Shall be performed anytime before screen 10.
2/ This is a stress test designed to insure a rugged product.

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4.3.1 Screening (JANC). Screening shall be in accordance with MIL-S-19500 (appendix H). As a minimum, die shall be 100 percent probed in accordance with group A.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein. Alternate flow is allowed for quality conformance inspection in accordance with figure 2 of MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. (End-point electrical measurements shall be in accordance with the applicable steps of table III herein.)

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table III herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

a. Subgroup 3: Condition for temperature cycling is test condtion G, 100 cycles.

b. Subgroup 3: Condition for hermetic seal are as follows:

Fine leak: Test condition G or H

Gross leak: Test condition C or D

c. Subgroup 4: Not applicable.

d. Subgroup 5: Condition for accelerated steady state reverse bias are as follows:

Condition A: V_{DS} = rated, T_A = +175°C, t = 120 hours. Read and record $V_{BR(DSS)}$ (pre and post) at I_D = 1.0 mA (N-CH), I_D = -1.0 mA (P-CH). Read and record I_{DSS} (pre and post), in accordance with table III. (NOTE: A separate sample may be pulled for each test)

e. Subgroup 5: Condition for accelerated steady state gate stress are as follows:

Condition B: V_{GS} = rated, T_A = +175°C, t = 24 hours. (NOTE: A separate sample may be pulled for each test)

f. Subgroup 6: Condition for Thermal resistance, see 4.5.2.

4.4.2.2 Group B inspection, table IVb (JAN, JANTX and JANTXV of MIL-S-19500.

a. Subgroup 2: Condition for temperature cycling is test condtion G, 25 cycles.

b. Subgroup 3: Condition for hermetic seal are as follows:

Fine leak: Test condition G or H

Gross leak: Test condition C or D

c. Subgroup 3: Condition for steady state reverse bias are as follows:

Test condition A: V_{DS} = 80 percent of rated. T_A = +150°C, t = 160 hours. (NOTE: A separate sample may be pulled for each test)

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d. Subgroup 3: Condition for steady state gate stress are as follows:

Test condition B: V_{GS} = 80 percent of rated. T_A = +150°C, t = 24 hours. (NOTE: A separate sample may be pulled for each test)

e. Subgroups 5 and 6 not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table III herein:

4.4.3.1 Group C inspection, table V of MIL-S-19500.

a. Subgroup 2: Condition for terminal strength (tension) is test condition A, weight = 10 pounds, t = 15 s.

b. Subgroup 3: Condition for hermetic seal are as follows:

Fine leak: Test condition G or H

Gross leak: Test condition C or D

a. Subgroup 6: Condition for steady state reverse bias are as follows:

Test condition A: V_{DS} = 80 percent of rated. T_A = +150°C, t = 340 hours.

b. Subgroup 6: Condition for steady state gate stress are as follows:

Test condition B: V_{GS} = 80 percent of rated. T_A = +150°C, t = 96 hours.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. $R_{\theta JA}$ maximum = 90°C/W, for each die.

a. I_M measuring current	10 mA
b. I_H drain heating current	0.15 A
c. t_H heating time	Steady-state (see method 3161 of MIL-STD-750 for definition)
d. V_H drain-source heating voltage	15 V
e. t_{MD} measurement time delay	30 to 60 µs
f. t_{SW} sample window time	10 µs maximum

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal response	3101	See 4.5.3	$Z_{\theta JC}$			
Breakdown voltage, drain to source	3407	$-V_{GS}$ = 0 V, condition C	$V_{(BR)DSS}$			
N-CHANNEL		I_D = 1.0 mA dc		100		V dc
P-CHANNEL		I_D = -1.0 mA dc		-100		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$	$V_{GS(th)1}$			
N-CHANNEL		I_D = 0.25 mA dc		2.0	4.0	V dc
P-CHANNEL		I_D = -0.25 mA dc		-2.0	-4.0	V dc
Gate current	3411	V_{GS} = +20 V and -20 V, bias condition C, V_{DS} = 0	I_{GSS1}		±100	nA dc
Drain current	3413	V_{GS} = 0 V; V_{DS} = 80% of rated V_{DS}, bias condition C	I_{DSS1}			
N-CHANNEL					25	µA dc
P-CHANNEL					-25	µA dc
Static drain to source on-state resistance	3421	Pulsed (see 4.5.1), condition A, I_D = rated I_{D2} (see 1.3)	$R_{DS(on)1}$			
N-CHANNEL		V_{GS} = 10 V			0.7	Ω
P-CHANNEL		V_{GS} = -10 V			0.7	Ω
Forward voltage (source drain diode)	4011	Pulsed (see 4.5.1), V_{GS} = 0 V, I_D = rated I_{D1}	V_{SD}			
N-CHANNEL		V_{GS} = 10 V			1.5	V dc
P-CHANNEL		V_{GS} = -10 V			-5.5	V dc
Subgroup 3						
High temperature operation		$T_C = T_J$ = +125°C				
Gate current	3411	Bias condition C; V_{GS} = +20 V and -20 V; V_{DS} = 0 V	I_{GSS2}		±200	nA dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued						
Drain current	3413	Bias condition C, V_{GS} = 0 V				
N-CHANNEL		V_{DS} = 100% of rated V_{DS}	I_{DSS2}		1.0	mA dc
		V_{DS} = 80% of rated V_{DS}	I_{DSS3}		0.25	mA dc
P-CHANNEL		V_{DS} = 100% of rated V_{DS}	I_{DSS2}		-1.0	mA dc
		V_{DS} = 80% of rated V_{DS}	I_{DSS3}		-0.25	mA dc
Static drain to source on-state resistance	3421	Pulsed (see 4.5.1); I_D = rated I_{D1}	$R_{DS(on)3}$			
N-CHANNEL		V_{GS} = 10 V			1.4	Ω
P-CHANNEL		V_{GS} = -10 V			1.4	Ω
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$	$V_{GS(th)2}$			
N-CHANNEL		I_D = 0.25 mA dc		1.0		V dc
P-CHANNEL		I_D = -0.25 mA dc		-1.0		V dc
Low temperature operation		$T_C = T_J$ = -55°C				
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$	$V_{GS(th)3}$			
N-CHANNEL		I_D = 0.25 mA dc			5.0	V dc
P-CHANNEL		I_D = -0.25 mA dc			-5.0	V dc
Subgroup 4						
Switching time test	3472	I_D = rated I_{D1}; R_G = 24Ω (N-CH); R_G = 18Ω (P-CH); V_{DD} = 0.5 $V_{BR(DSS)}$				
Turn on delay time			$t_{d(on)}$			
N-CHANNEL		V_{GS} = 10 V			20	nS dc
P-CHANNEL		V_{GS} = -10 V			30	nS dc
Rise time			t_r			
N-CHANNEL		V_{GS} = 10 V			25	nS dc
P-CHANNEL		V_{GS} = -10 V			60	nS dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Turn off delay time			$t_{d(off)}$			
N-CHANNEL		V_{GS} = 10 V			40	nS dc
P-CHANNEL		V_{GS} = -10 V			60	nS dc
Fall time			t_f			
N-CHANNEL		V_{GS} = 10 V			40	nS dc
P-CHANNEL		V_{GS} = -10 V			60	nS dc
Subgroup 5						
Single pulse avalanche energy	3470	See (4.5.4)	E_{AS}			
Electrical measurements		See table III herein, steps 1, 2, 3, 4, 5, and 6				
Safe operating area test	3474	See figure 4, V_{DS} = 80% of rated $V_{BR(DSS)}$, tp = 10 ms				
N-CHANNEL		I_D = 0.25 A				
P-CHANNEL		I_D = -0.25 A				
Electrical measurements		See table III herein, steps 1, 2, 3, 4, 5, 6, and 7				
Subgroup 6						
Not applicable						
Subgroup 7						
On state gate charge	3471	Condition B	$Q_{g(on)}$			
N-CHANNEL					15	nC dc
P-CHANNEL					22	nC dc
Gate to source charge			Q_{gs}			
N-CHANNEL					7.5	nC dc
P-CHANNEL					8.0	nC dc
Gate to drain charge			Q_{gd}			
N-CHANNEL					7.5	nC dc
P-CHANNEL					14.0	nC dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 7 - Continued						
Reverse recovery time	3473	di/dt ≤ 100 A/μS, I_D = rated I_{D1}	t_{rr}			
N-CHANNEL		V_{DD} ≤ 30 V			200	nS dc
P-CHANNEL		V_{DD} ≤ -30 V			200	nS dc

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750	
	Method	Conditions
Subgroup 1		
Temperature cycle	1051	Test condition G, 500 cycles
Hermetic seal		
Fine leak		Test conditions G or H
Gross leak		Test conditions C or D
Electrical measurements		See table III herein, steps 1, 2 3, 4, 5, 6, and 7
Subgroup 2		
Steady-state reverse bias	1042	Condition A, 1,000 hours
Electrical measurements		See table III herein, steps 1, 2, 3, 4, 5, and 6
Steady-state gate bias	1042	Condition B, 1,000 hours
Electrical measurements		See table III herein, steps 1, 2, 3, 4, 5, and 6
Subgroup 3		
Destructive physical analysis (DPA)	2102	
Subgroup 4		
Thermal resistance	3161	$R_{\theta JA}$ = 90°C/W maximum, see 4.5.2
Subgroup 5		
Not applicable		

1/ For sampling plan, see MIL-S-19500.

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TABLE III. Groups A, B, C, and E electrical end-point measurements.

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Breakdown voltage drain to source	3407	V_{GS} = 0 V, bias condition C	$V_{(BR)DSS}$			
	N-CHANNEL		I_D = 1.0 mA dc		100		V dc
	P-CHANNEL		I_D = -1.0 mA dc		-100		V dc
2.	Gate to source voltage (threshold)	3404	$V_{DS} \geq V_{GS}$,	$V_{GS(th)1}$			V dc
	N-CHANNEL		I_D = 0.25 mA dc		2.0	4.0	V dc
	P-CHANNEL		I_D = -0.25 mA dc		-2.0	-4.0	V dc
3.	Gate current	3411	V_{GS} = ±20 V dc, bias condition C	I_{GSS1}		±100	nA dc
4.	Drain current	3413	V_{GS} = 0, V_{DS} = 80% of rated V_{DS}, bias condition C	I_{DSS1}			
	N-CHANNEL					25	µA dc
	P-CHANNEL					-25	µA dc
5.	Static drain to source on-state resistance	3421	Condition A, pulsed (see 4.5.1), $I_D = I_{D2}$	$R_{DS(on)1}$			
	N-CHANNEL		V_{GS} = 10 V			0.7	Ω
	P-CHANNEL		V_{GS} = -10 V			0.7	Ω
6.	Forward voltage (source-drain diode)	4011	Pulsed (see 4.5.1), V_G = 0 V, $I_D = I_{D1}$	V_{SD}			
	N-CHANNEL					1.5	V dc
	P-CHANNEL					-5.5	V dc
7.	Thermal response	3161	See 4.5.3	$Z_{\theta JC}$			

See footnotes at end of table.

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TABLE III. Groups A, B, C, and E electrical end-point measurements - Continued.

1/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:

a. Subgroup 3, see table III herein, steps 1, 2, 3, 4, 5, 6, and 7.

b. Subgroup 5, see table III herein, steps 1, 2, 3, 4, 5, and 6. (No more than 15 percent of the sample shall be permitted to have a $\Delta V_{BR(DSS)}$ shift of more than 10 percent and ΔI_{DSS} greater than 50μA)

2/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table III herein, steps 1, 2, 3, 4, 5, 6, and 7.

b. Subgroup 3, see table III herein, steps 1, 2, 3, 4, 5, and 6.

3/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroup 2, see table III herein, steps 1, 2, 3, 4, 5, 6, and 7 for all levels.

b. Subgroup 3 and 6, see table III herein, steps 1, 2, 3, 4, 5, and 6 for all levels.

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4.5.3 Thermal response ($Z_{\theta JC}$) measurements). The $Z_{\theta JC}$ measurements shall be performed in accordance with method 3161 of MIL-STD-750. The maximum limit for $Z_{\theta JC}$ is screening (Table II of MIL-S-19500) shall be derived by each vendor by means of Statistical Process Control (SPC). When three (minimum) lot date codes have exhibited control, the data from these three minimum lot date codes will be used to establish a fixed screening limit, (not to exceed the group A sub 2 limit). Once a fixed limit has bead and record) in the qualification report. The chosen ΔV_{SD} shall be considered final after the manufacturer has had the opportunity to test five consecutive lots. The following parameter measurements shall apply:

a.	I_M measuring current	10 mA
b.	I_H drain heating current	0.15 A minimum
c.	t_H heating time	100 ms
d.	V_H drain-source heating voltage	15 V minimum
e.	t_{MD} measurement time delay	30 to 60 μs
f.	t_{SW} sample window time	10 μs maximum

4.5.4 Single pulse avalanche energy E_{AS}.

a.	Peak current (I_{AS})	I_{D1}
b.	Peak gate voltage (V_{GS})	10 V
c.	Gate to source resistor (R_{GS})	$25 \leq R_{GS} \leq 200\Omega$
d.	Initial case temperature	+25°C +10°C, -5°C
e.	Inductance	$[2E_{AS}/(I_{D1})^2]$ $[\frac{(V_{BR} - V_{DD})}{V_{BR}}]$ mH minimum
f.	Number of pulses to be applied	1 pulse minimum
g.	Supply voltage (V_{DD})	25 V (N-Channel); -25 V (P-Channel)

4.5.5 Gate stress test.

a. V_{GS} = 30 V minimum.

b. t = 250 μs minimum.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation (see 2.1).
b. Lead finish specified (see 3.3.1).
c. Product assurance level and type designation.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation (see 2.1).
b. Lead finish specified (see 3.3.1).
c. Product assurance level and type designation.

6.3 Substitution information. Devices covered by this specification are substitutable for the manufacturers' and users' Part or Identifying Number (PIN). This information in no way implies that manufacturers' PIN's are suitable as a substitute for the military PIN.

Military PIN	Manufacturers' CAGE code	Manufacturers' and users' PIN
2N7337	59993	IRFG5110

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Air Force - 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC, OS
Air Force - 13

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5961-1382)

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